

Computer Networking Homework #1

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1. Suppose Host A wants to send a large file to Host B. The path from Host A to Host B contains three hops (links), with the link rates of these hops of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps. That is, the file sent by Host A must pass through two immediate routers to reach Node B.

- (a) Assuming no other traffic in the network, what is the throughput for the file transfer.

A: 500 kbps

- (b) Repeat (a), but now with R_3 reduced to 200 Kbps.

A: 200 kbps

2. Suppose users share a 2 Mbps link. Also suppose each user requires 100 Kbps when transmitting, but each user transmits only 20% of the time in average. (You may review the concept of statistical multiplexing for solving this problem)

- (a) When circuit switching is used, how many users can be supported?

$$(2 \times 10^6) \div (100 \times 10^3) = 20 \#$$

A: 20人

- (b) For the remainder of this problem, suppose packet switching is used. Find the probability, p , that a given user is transmitting.

$$20\% = 0.2$$

A: 0.2

- (c) Now suppose there are 40 users. Then the probability that at any given time, exactly n users are transmitting simultaneously is formulated as the binomial distribution of $\binom{40}{n} p^n (1-p)^{40-n}$, where p is the answer from part (b).
Therefore, if more than 20 users are transmitting simultaneously using packet switching, the queuing situation or packet loss may occur. Under this condition, write the formula of the probability: P(21 or more users transmitting simultaneously).

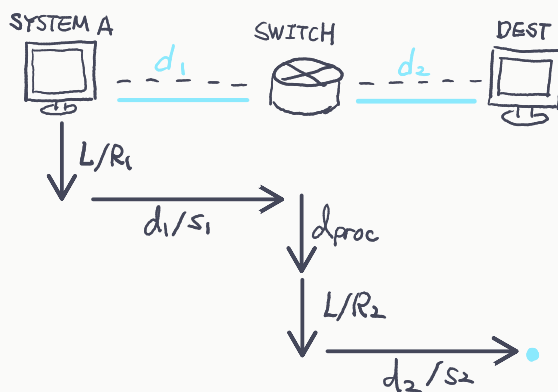
$$p = 0.2$$

$$A: \sum_{n=21}^{40} \binom{40}{n} (0.2)^n (0.8)^{(40-n)}$$

3. Consider the queuing delay in a router buffer (preceding an outbound link). Suppose all packets are L bits, the transmission rate is R bps and that N packets simultaneously arrive at the buffer every LN/R seconds, where $R \gg LN$. Find the average queuing delay of a packet. (Hint: The queuing delay for the first packet is zero; for the second packet L/R ; for the third packet $2L/R$. The N th packet has already been transmitted when the second batch of packets arrives.)

$$\begin{aligned} \text{Total delay time} &= \sum_{i=1}^N \frac{Li}{R} \\ &= \frac{L}{R} \sum_{i=1}^N ix = \frac{L}{R} \sum_{i=1}^N i \\ &= \frac{L}{R} \times \frac{(1+n) \times n}{2} = \frac{L \times (n+n^2)}{2R} \end{aligned} \quad \begin{aligned} \text{Average Delay} &= \frac{L \times (n+n^2)}{2R} \times \frac{1}{n} \\ &= \frac{L \times (n+1)}{2R} \\ \text{A: } &\frac{L \times (n+1)}{2R} \end{aligned}$$

4. Consider a packet of length L bits which begins at end system A, travels over one link to a packet switch, and travels from the packet switch over a second link to a destination end system. Let d_i , s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2$. The packet switch processes and delays each packet by d_{proc} . Assuming no queuing delays, in terms of d_i , s_i , R_i , ($i = 1, 2$), and L , what is the total end-to-end delay for the packet? Suppose now the packet length is 1,000 bytes, the propagation speed on both links is $2 \cdot 10^8$ m/s, the transmission rates of both links is 2 Mbps, the packet switch processing delay is 1 msec, the length of the first link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?



$$\frac{L}{R_1} = \frac{1000 \times 8}{2 \times 10^6} = 5 \times 10^{-4} \times 8 = 4 \text{ ms}$$

$$\frac{L}{R_2} = \frac{1000 \times 8}{2 \times 10^6} = 4 \text{ ms}$$

$$\frac{d_1}{s_1} = \frac{4 \times 10^6}{2 \times 10^8} = 2 \times 10^{-2} = 20 \text{ ms}$$

$$\frac{d_2}{s_2} = \frac{1 \times 10^6}{2 \times 10^8} = 0.5 \times 10^{-2} = 5 \text{ ms}$$

$$d_{\text{proc}} = 1 \text{ ms}$$

$$\text{delay} = 4 + 4 + 20 + 5 + 1 = 34 \text{ (ms)}$$

$$\text{A: (1) } \frac{L}{R_1} + \frac{L}{R_2} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + d_{\text{proc}}$$

(2) 34 ms #

5. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

A: Five layers are:

Application layer: Supporting network applications e.g. HTTP, IMAP

Transport layer: process to process data transfer e.g. TCP, UDP

Network layer: routing data from source to destination e.g. IP

Link layer: data transfer between neighboring network element e.g. Ethernet

Physical layer: bits on the wire

6. Explain briefly what encapsulation and decapsulation mean in the layered protocol stack.

A: Encapsulation means that tags are added to the head of the data from each layer, makes router to know how to forward it.

Decapsulation means that read the tags in front of the data and removing it, then shows the original data to the top layer eventually.